

Arjuna JEE 2025

Maths

DPP: 8

Trigonometric Identities

- Q1** If $A + B + C = \pi$, then $\sin 2A + \sin 2B + \sin 2C$ is equal to
 (A) $4 \sin A \sin B \sin C$
 (B) $4 \cos A \cos B \cos C$
 (C) $2 \cos A \cos B \cos C$
 (D) $2 \sin A \sin B \sin C$
- Q2** If $A + B + C = \pi$, then $\cos A + \cos B - \cos C = \left(4 \cos \frac{A}{2} \cdot \cos \frac{B}{2} \sin \frac{C}{2}\right) - m$, then value of m is
 (A) 0 (B) 1
 (C) -1 (D) 2
- Q3** $\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11}$ is equal to
 (A) 1
 (B) $\frac{1}{2}$
 (C) $\frac{1}{4}$
 (D) 2
- Q4** $\sin \theta + \sin 3\theta + \sin 5\theta + \dots \sin(2n-1)\theta$ is equal to
 (A) $\operatorname{cosec} \theta - \frac{\cos^2 n\theta}{\sin \theta}$
 (B) $\frac{\sin n\theta}{\sin \theta}$
 (C) $\sin n\theta$
 (D) $\cos^2 n\theta$
- Q5** $\sum_{r=1}^{n-1} \cos^2 \frac{r\pi}{n}$ is equal to
 (A) $\frac{n}{2}$
 (B) $\frac{n-1}{2}$
 (C) $\frac{n}{2} - 1$
 (D) $\frac{n+1}{2}$
- Q6** The value of $\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$ to n terms is equal to
 (A) 1
 (B) 0
 (C) $\frac{n}{2}$
 (D) $\frac{n-1}{2}$
- Q7** The value of $\cos\left(\frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{\pi}{16}\right) \dots \cos\left(\frac{\pi}{2^n}\right)$ equals
 (A) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$
 (B) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$
 (C) $\frac{1}{2^n} \operatorname{cosec}\left(\frac{\pi}{2^{n-1}}\right)$
 (D) $\frac{1}{2^{n-1}} \operatorname{cosec}\left(\frac{\pi}{2^n}\right)$
- Q8** The expression $2 \sin 2^\circ + 4 \sin 4^\circ + 6 \sin 6^\circ + \dots + 180 \sin 180^\circ$ equals
 (A) $\cot 1^\circ$
 (B) $90 \cot 1^\circ$
 (C) $\sin 1^\circ$
 (D) $90 \cos 1^\circ$
- Q9** If $\tan \alpha = (1 + 2^{-x})^{-1}$, $\tan \beta = (1 + 2^{x+1})^{-1}$, then $\alpha + \beta =$
 (A) $\pi/6$ (B) $\pi/4$
 (C) $\pi/3$ (D) $\pi/2$
- Q10** If the solution of $a \cos 2\theta + b \sin 2\theta = c$ is α and β then the value of $\tan \alpha + \tan \beta$ is
 (A) $\frac{c+a}{2b}$
 (B) $\frac{2b}{c+a}$



- (C) $\frac{c-a}{2b}$
 (D) $\frac{b}{c+a}$

Q11 Let $S = \sum_{r=1}^5 \cos(2r-1)\frac{\pi}{11}$ and $P = \prod_{r=1}^4 \cos(2^r \frac{\pi}{15})$, then
 (A) $\log_s P = -4$
 (B) $P = 3S$
 (C) $\operatorname{cosec} S > \operatorname{cosec} P$
 (D) $\tan^{-1} P < \tan^{-1} S$

Q12 If $\theta = \frac{\pi}{2^{n+1}}$, then the value of $\cos \theta \cos 2\theta \cos 2^2\theta \dots \cos 2^{n-1}\theta$ is
 (A) 1
 (B) $1/2^n$
 (C) 2^n
 (D) none of these

Q13 Given that $\tan A$ and $\tan B$ are the roots of the equation $x^2 - ax + b = 0$. The value of $\sin^2(A+B)$, is
 (A) $\frac{a^2}{a^2+(1-b)^2}$
 (B) $\frac{a^2}{a^2+b^2}$
 (C) $\frac{a^2}{(a+b)^2}$
 (D) $\frac{a^2}{b^2+(1-a)^2}$

Q14 If $\sin A = n \sin B$, then $\frac{n-1}{n+1} \tan \frac{A+B}{2} =$
 (A) $\sin \frac{A-B}{2}$
 (B) $\tan \frac{A-B}{2}$
 (C) $\cot \frac{A-B}{2}$
 (D) None of these

Q15 Which of the following is 0 at $x = n\pi, n \in I$
 (A) $\tan x$ (B) $\sin x$
 (C) $\cot x$ (D) $\cos x$



Answer Key

Q1 (A)
Q2 (B)
Q3 (B)
Q4 (A)
Q5 (C)
Q6 (B)
Q7 (D)
Q8 (B)

Q9 (B)
Q10 (B)
Q11 (D)
Q12 (B)
Q13 (A)
Q14 (B)
Q15 (A, B)



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